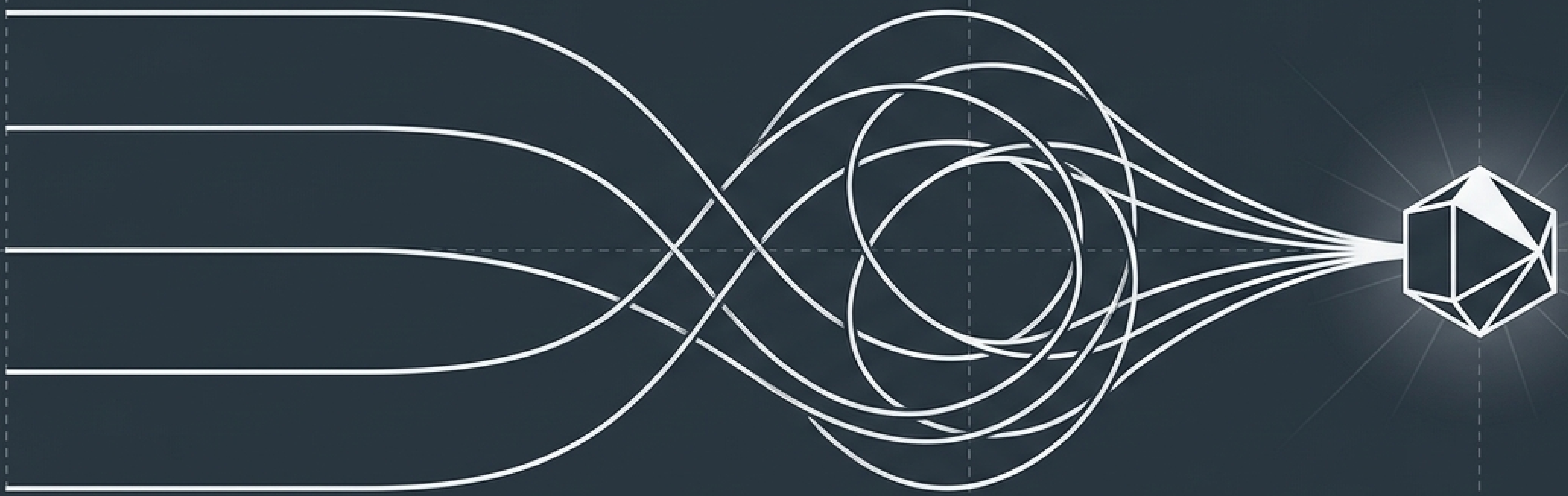
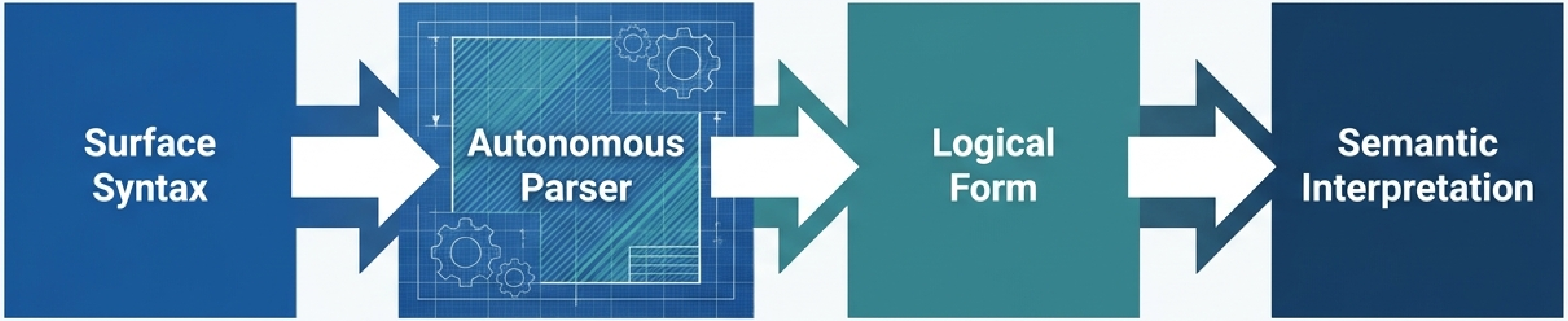


What Survives Reconstruction

Admissibility, Irreversibility, and the Architecture of Externalized Structure

Flyxion | July 2026

The Category Mistake of the Claude Code Leak



The Incident

In March 2026, the complete source of Anthropic's Claude Code CLI leaked, revealing hand-written deterministic parsers alongside the neural model.

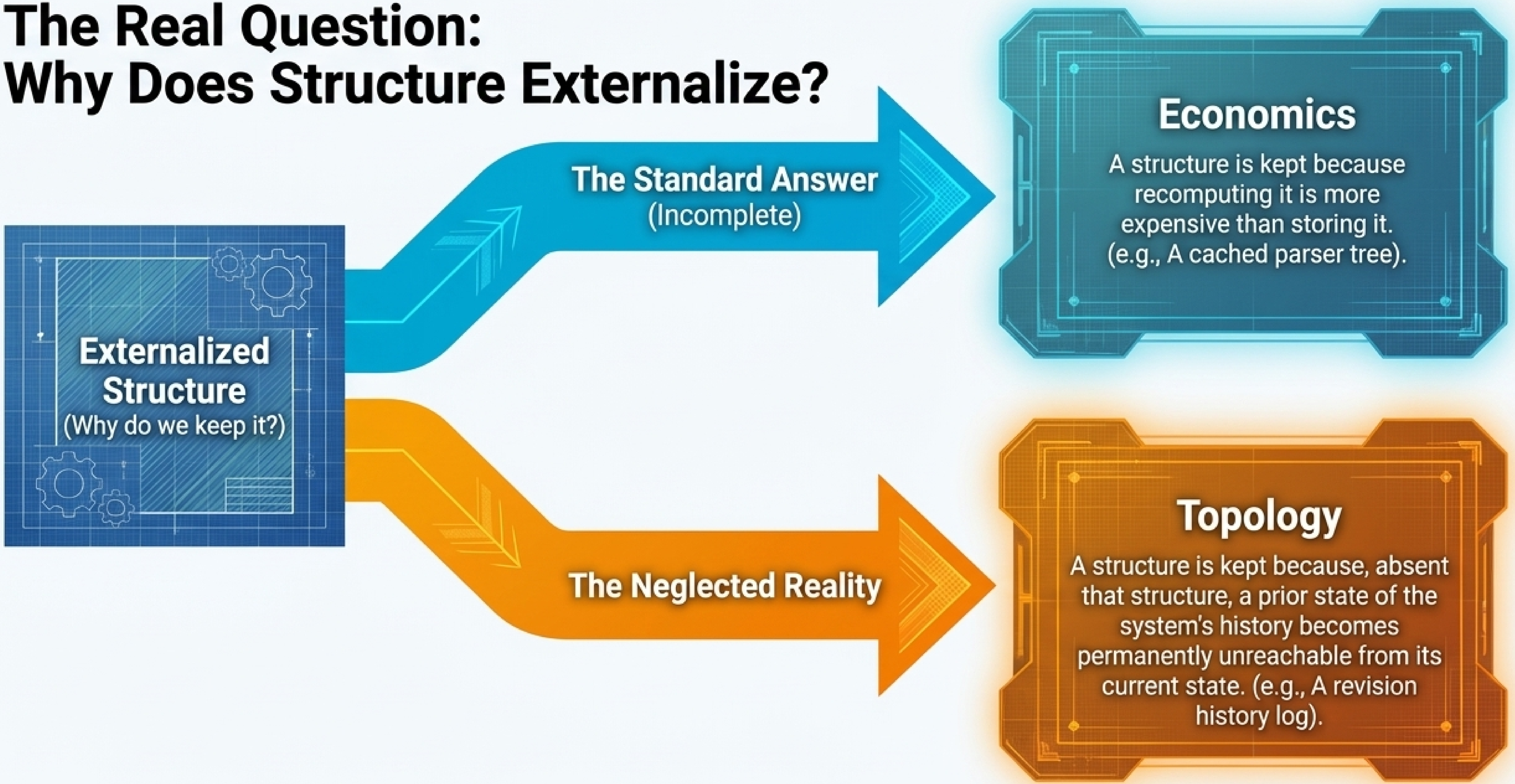
The False Inference

The public claimed finding a non-neural parser proved the AI possessed no real understanding—an illusion of a wizard behind a curtain.

The Rebuttal

Generative linguistics established long ago that syntactic computation possesses explanatory autonomy. Finding a parser is evidence of mature engineering, not a lack of semantic competence.

The Real Question: Why Does Structure Externalize?



The Two Kinds of Persistence

Reconstruction-Cost Materialization	Irreversibility Materialization
Driver: Computational Economics.	**Driver:** System Topology.
Condition: A Generating Input exists (a state independent of execution history from which the structure can be reproduced, e.g., a source file on disk).	**Condition:** No Generating Input exists. The transition is Reconstructively Irreversible.
Contingency: Tracks the cost curve. Could disappear as compute gets cheaper.	**Contingency:** Absolute necessity. No improvement in hardware speed dissolves this requirement, because it is a claim about history, not cost.

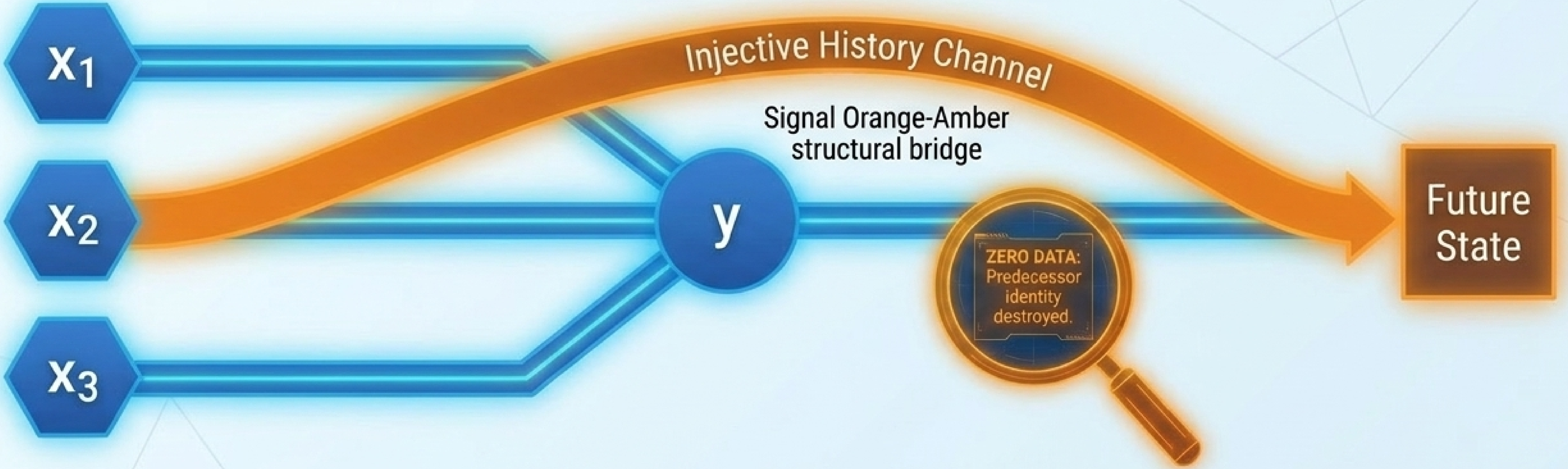
Mechanics of Information Loss: Fiber Collapse

Transition System
A system defined by a transition function. Editing a file, revising a belief, or compacting context.

Fiber Collapse
Non-injectivity is the ordinary case—transitions generically collapse fibers. Indistinguishable inputs yield a single output.

Proposition 4.6
If a function is non-injective over a fiber, no function of the successor state **y alone determines which predecessor produced it. The predecessor is not just expensive to reconstruct. It is destroyed.**

Formalizing Irreversibility: History Channels



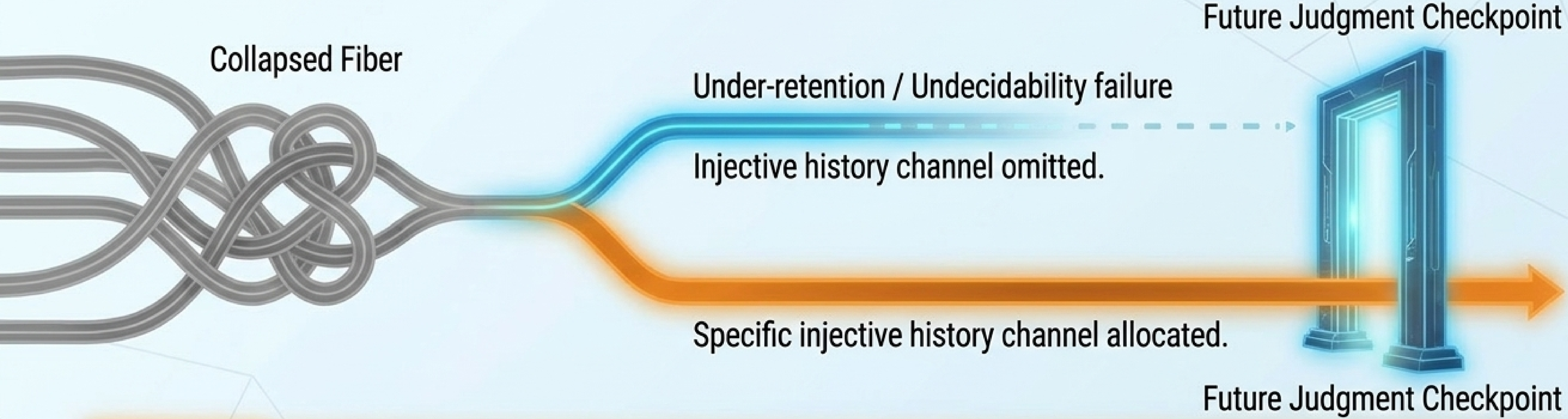
Historical Distinction Theorem (Theorem 4.9)

- To repair a collapsed fiber, a system augments the state with a history channel.
- The identity of the predecessor is recoverable from the current state if and only if the history channel is injective over the specific fiber that collapsed.

The Trap of Mere Retention

- A log recording “a file was edited” is non-injective (useless for recovery).
- A log recording the diff is injective.
- The mere presence of a trace does not entail it preserves distinguishability.

The Selective Retention Theorem: What Actually Matters



- A system cannot preserve distinguishability over every non-injective transition. It would drown in its own history.
- **Admissibility-Relevant Fibers:** A fiber matters only if some future admissibility judgment depends on which predecessor actually occurred.
- **Theorem 5.10 (Selective Retention):** A system minimizes retention cost by allocating injective history channels precisely over admissibility-relevant fibers, and discarding distinguishability everywhere else.

The Falsifiability Audit: How to Test the Theory

1. Enumerate Candidate Fibers

Inspect system transition dynamics for non-injective operations (edits, compactions, merges).

2. Test Relevance Externally

Replay system dynamics from distinct predecessors.
If a downstream admissibility verdict diverges, the fiber is admissibility-relevant.

3. Audit Retention

Independently inspect if the system retains a structure that is injective over that specific fiber.

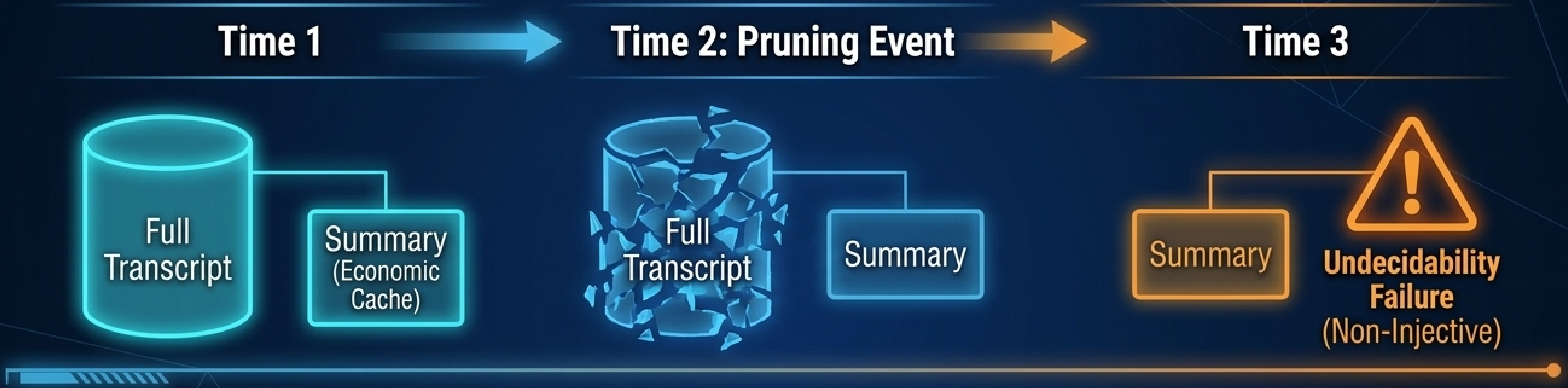
4. Compare & Diagnose

- Relevant + Retained = Compliant
- Relevant + Not Retained = Under-Retention (Undecidability Failure)
- Not Relevant + Retained = Over-Retention (Wasted Resources)

Case Study: Auditing the Claude Code Leak

Component	Generating Input?	Admissibility-Relevant?	Retention Observed?	Final Classification
Parse/Search Infrastructure	Yes (source disk)	No	Cached	Reconstruction-Cost
Pre-Risky-Operation Checkpoint	Yes (explicitly created)	Yes (rollback depends on it)	Retained	Irreversibility
Context Compaction Chain	Yes (while transcript lives)	Yes	Retained, boundaries patched	Ambiguous (Time-Dependent)
Cross-Session Trust State	No	Contested	Not Retained	Failure Mode Identified

The Decay of Generating Inputs: Checkpoints vs. Compaction



Confirmation (Checkpoints)

Checkpoints strictly confirm the theorem. Triggered specifically before risky operations, deliberately aligning retention with admissibility-relevance.

Ambiguity (Context Compaction)

A summarized context is Reconstruction-Cost only as long as the raw transcript survives.

The Silent Reclassification

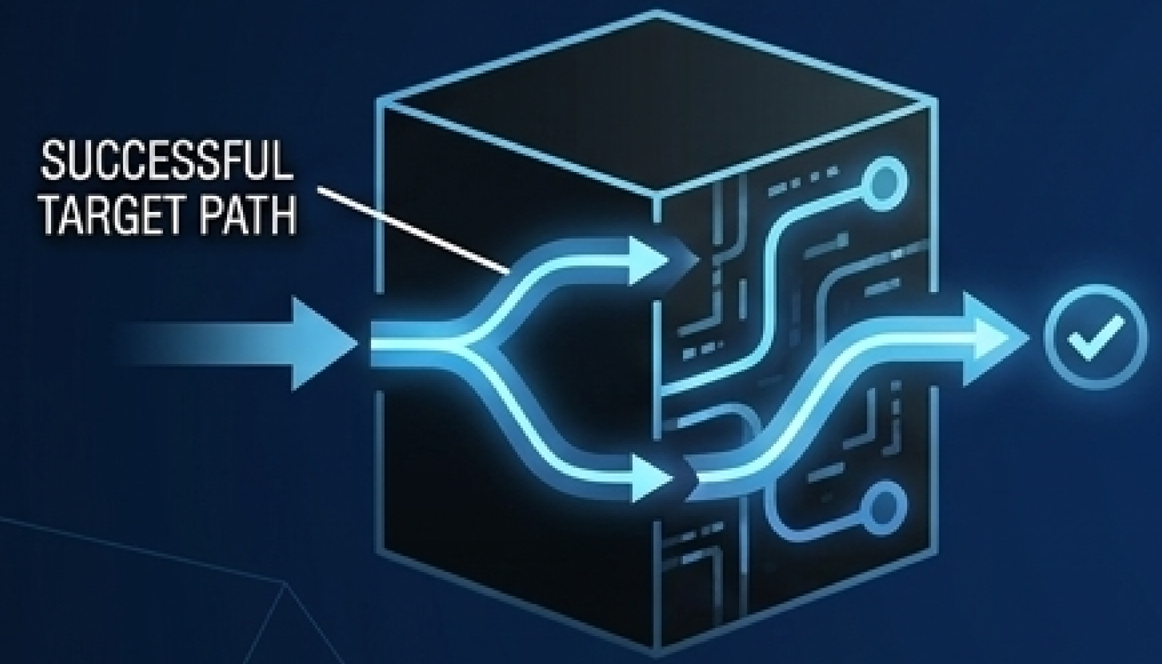
When memory limits prune the raw transcript, the summary silently becomes Irreversibility Materialization.

Because the summary is inherently lossy (non-injective), it fundamentally fails to preserve admissibility, causing silent undecidability failures downstream.

The Transformer Semantics Reframing

The “Logical Form” debate conflated two entirely different questions.

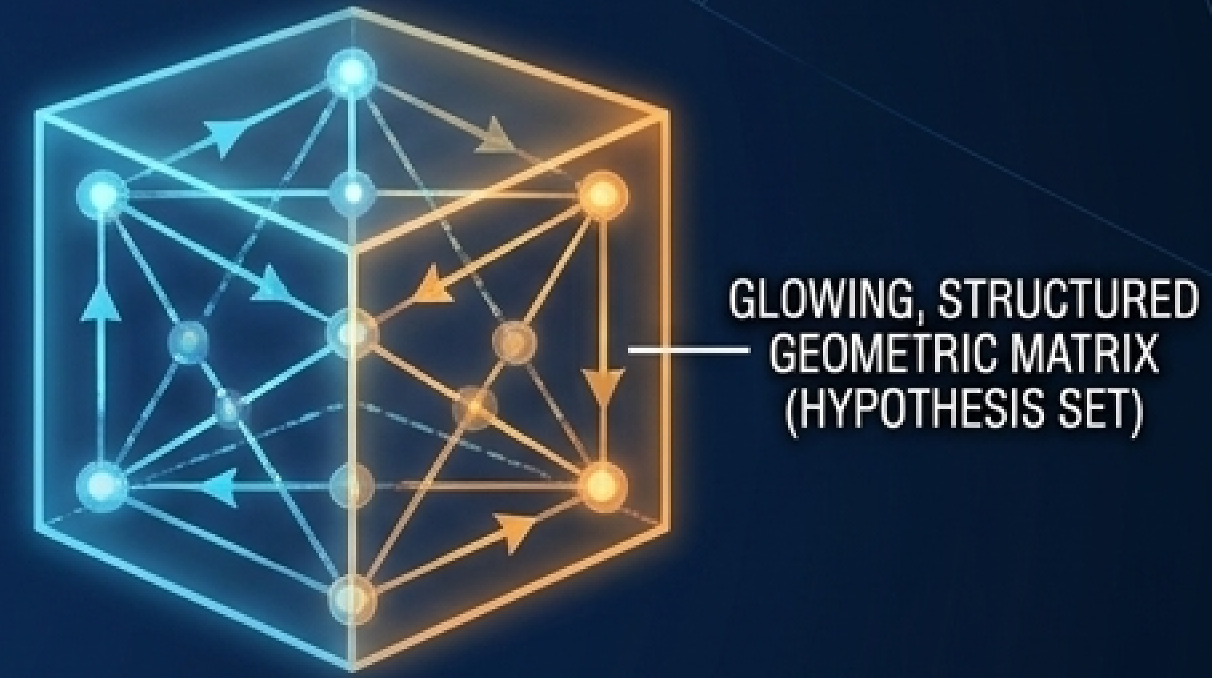
1. Functional Adequacy



The Test: Do downstream outputs track the true predecessor at better than chance across counterfactual contexts (e.g., distinguishing quantifier scopes)?

The Meaning: The system behaviorally resolves the collapsed fiber.

2. Architectural Transparency



The Test: Does the system maintain an explicit, inspectable object (a hypothesis set) whose state can be compared against the fiber without probing latent vectors?

The Meaning: The system is engineered to track its own admissibility, matching HYDRA-level explicit representation. A model can be functionally adequate without being architecturally transparent.

The Physics of Persistence

Hamiltonian Mechanics

Primitive:
Instantaneous State
(q, p)

Question:
What can be recovered from a state?

Dynamics:
Flow is reversible (never collapses fibers).

Lagrangian Mechanics

Primitive:
Whole Trajectory

Question:
Which trajectory is optimal?

Dynamics:
Extremal selection.

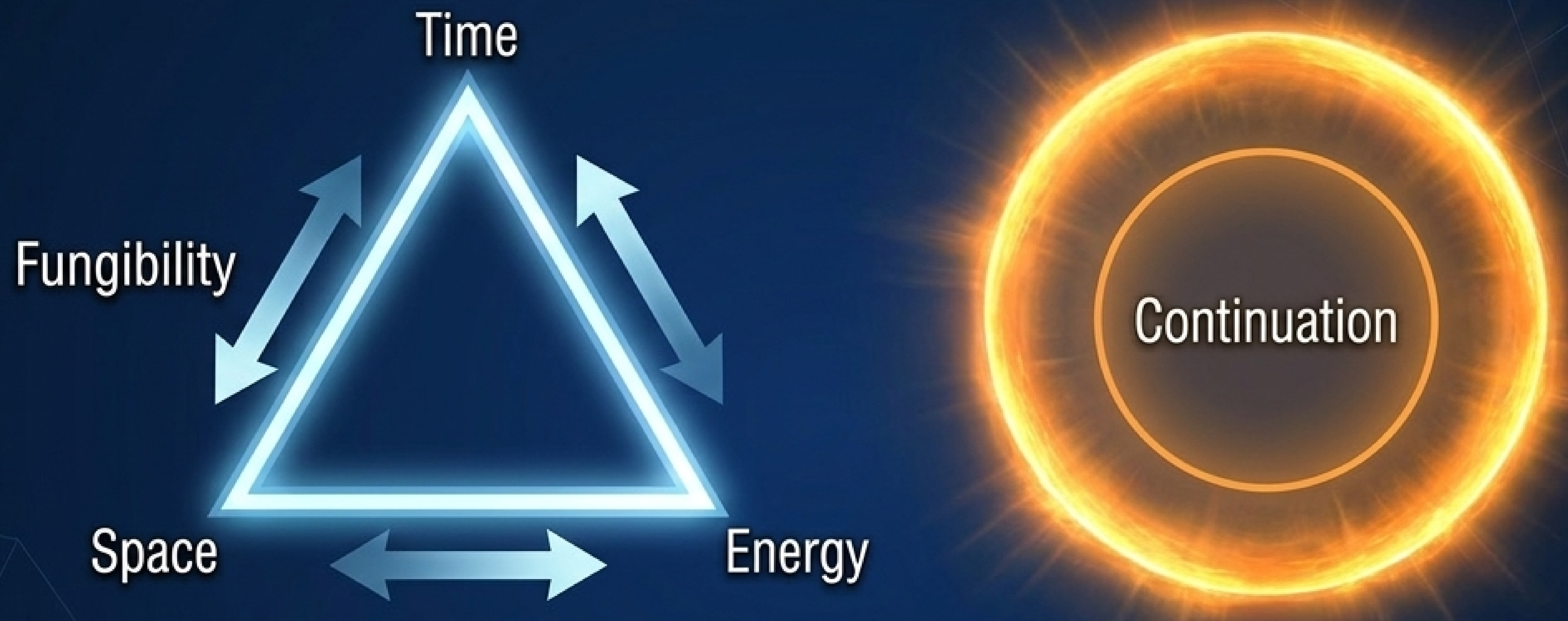
Continuation Geometry

Primitive:
Admissibility relative to a continuation family

Question:
What survives reconstruction?

Dynamics:
Generic fiber collapse, persistence secured via repair under anomaly, not optimization.

Continuation as the Fourth Computational Resource

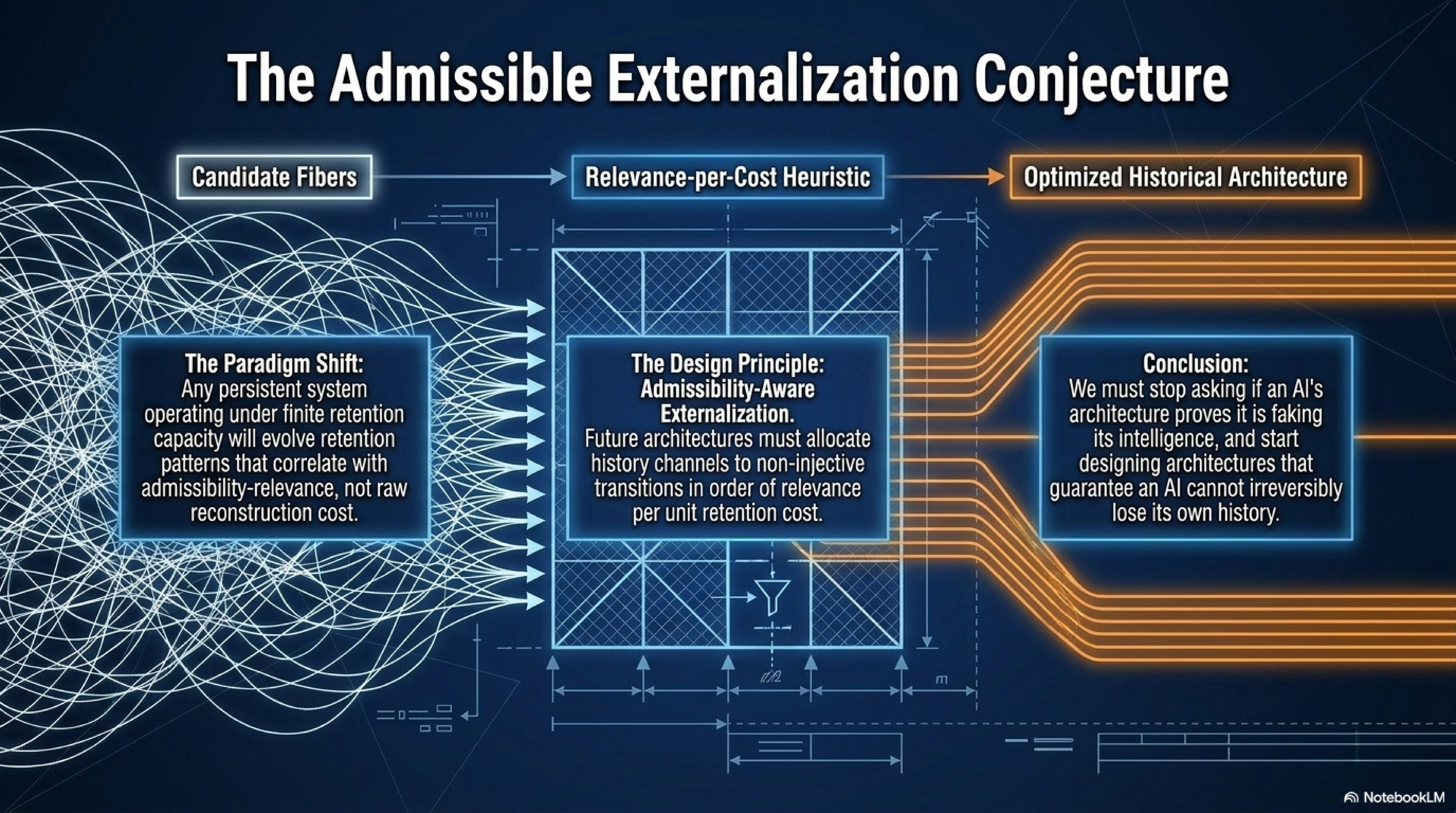


Time, Space, and Energy are fungible. You can trade memory for compute, or energy for speed.

The **Resource Independence Theorem**: If a generating input is destroyed, and the transition was reconstructively irreversible, no allocation of Time, Space, or Energy can recover the predecessor's identity.

Continuation is not "more memory." Unbounded memory cannot save a system that applied a non-injective retention policy.

Historical Observability is an independent dimension of computation.


The Admissible Externalization Conjecture
Candidate Fibers
Relevance-per-Cost Heuristic
Optimized Historical Architecture
The Paradigm Shift:
Any persistent system operating under finite retention capacity will evolve retention patterns that correlate with admissibility-relevance, not raw reconstruction cost.
The Design Principle: Admissibility-Aware Externalization.
Future architectures must allocate history channels to non-injective transitions in order of relevance per unit retention cost.
Conclusion:
We must stop asking if an AI's architecture proves it is faking its intelligence, and start designing architectures that guarantee an AI cannot irreversibly lose its own history.
NotebookLM